

Ethan Shen

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Education

University of Washington

PhD in Computer Science and Engineering

Advisor: Ali Farhadi

Seattle, WA

Sep. 2025 - Jun. 2028

University of Washington

B.S. in Computer Science, Minor in Mathematics (GPA: 3.96/4.00)

Magna Cum Laude

Seattle, WA

Aug. 2022 - Jun. 2025

Publications

SERA: Soft-Verified Efficient Repository Agents

Ethan Shen, Daniel Tormoen, Saurabh Shah, Ali Farhadi, Tim Dettmers.
Under Review.

Perception Tokens Enhance Visual Reasoning in Multimodal Language Models

Mahtab Bigverdi, Zelun Luo, Cheng-Yu Hsieh, **Ethan Shen**, Dongping Chen, Linda Shapiro, Ranjay Krishna.
CVPR 2025.

Superposed Decoding: Multiple Generations from a Single Autoregressive Inference Pass

Ethan Shen, Alan Fan, Sarah M Pratt, Jae Sung Park, Matt Wallingford, Sham M Kakade, Ari Holtzman, Ranjay Krishna, Ali Farhadi, Aditya Kusupati.
NeurIPS 2024.

Are “Hierarchical” Visual Representations Hierarchical?

Ethan Shen, Ali Farhadi, Aditya Kusupati.
Workshop on Symmetry and Geometry in Neural Representations @ NeurIPS 2023.

Research Experience

Allen Institute of Artificial Intelligence

Research Intern (Advisor: Prof. Tim Dettmers)

Seattle, WA

Feb. 2025 - Present

- Leading development of Ai2’s coding agents starting with SERA, a SoTA open-source coding agent with 8B/32B parameters.
- Experimented with network stacking for low-cost LLM scaling and implemented distributed training for 100M to 7B parameter models on a trillion-token scale.

Reasoning, AI, Vision Lab @ UW

Research Assistant (Advisors: Prof. Ali Farhadi, Prof. Ranjay Krishna)

Seattle, WA

Jun. 2023 - Present

- Pioneered latent token representations for multimodal reasoning in vision-language models like Qwen and LLaVA.
- Created LLM decoding algorithm to improve inference efficiency and accuracy for multi-draft generation.
- Developed suite of hierarchical vision datasets for understanding complex visual hierarchies.

Sensor Systems Lab @ UW

Research Assistant (Advisor: Prof. Joshua Smith)

Seattle, WA

Jun. 2022 - Jun. 2023

- Designed and built an acoustic levitator, a tool that uses ultrasonic sound to levitate fragile objects for scientific experiments.
- Implemented dynamics simulations in Python and C++ to predict and control the rotation of objects in the levitator, with a 15x speedup over existing simulations.

Select Media Coverage

GeekWire – “Ai2 Cooks Up Open-Source Coding Agents With Tech Equivalent Of A Hot Plate And A Frying Pan” [Link](#)

AI Business – “Ai2 Releases Open Coding Agents Family” [Link](#)

SiliconANGLE – “Ai2 Launches Family Of Open-Source AI Developer Agents That Adapt To Codebases” [Link](#)

Nvidia AI – “Congratulations to Ai2 Open Coding Agents” [Link](#)

Awards & Fellowships

2026 **Fellow**, Paul and Daisy Soros Fellowship for New Americans

2026 **Fellow**, NSF GRFP Fellowship

2023 **NeurIPS Travel Award**, Conference for Neural Information Processing Systems

2022 **FEEA Merit Scholarship**, US Department of Veterans Affairs

2021 **National Merit Finalist**, National Merit Program

Work Experience

- Amazon

Seattle, WA

Software Engineer Intern

Jun. 2024 - Sep. 2024

 - Built a scalable GenAI pipeline with AWS Bedrock, Lambda, and DynamoDB to automatically process up to 15,000 financial documents a month, saving over 1200 hours of manual work per year.
 - Implemented fine tuning and heuristic filtering to achieve 96% accuracy on an internal invoice dataset.
- Papyrus

San Francisco, CA

Machine Learning Engineer Intern

Dec. 2023 - Mar. 2024

 - Worked at pre-seed startup and created novel feature to label speakers in transcripts using long-context LLMs.
 - Developed ASR pipeline for transcription/translation with Whisper and improved diarization with audio processing tools like Demucs and Pyannote.

Professional Services

- 2026-Now

ICML Reviewer (2026)
- 2025-Now

NeurIPS Reviewer (2025, 2026)
- 2024-Now

EMNLP Reviewer (2024, 2025)
- 2023-Now

UW CSE: Teaching Assistant
- 2023-2024

UW CSE: Student Interviewer for Faculty Hiring
- 2022-2023

UW CSE: Student Advisory Council Officer

Presentations

- Oct. 2024

UW RAIVN Lab (Host: Ali Farhadi), Topic: Superposed Decoding
- Aug. 2024

AWS AI Labs CodeGen (Host: Zijian Wang), Topic: Superposed Decoding
- Aug. 2024

Amazon LLM Reasoning Group (Host: Linbo Liu), Topic: Superposed Decoding
- Jul. 2024

AWS AI Labs Quicksight (Host: Patrick Ng), Topic: Superposed Decoding

References

- Ali Farhadi

Professor, University of Washington

Corporate Vice President, Microsoft Superintelligence
- Tim Dettmers

Assistant Professor, Carnegie Mellon University

Research Scientist Manager, Allen Institute of AI
- Aditya Kusupati

Staff Research Scientist, Google Deepmind